Lec 06-07
Communication models

Computer Networks (CSE232)

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These slides are inspired and adapted from:

1. *Computer Networking: A Top-Down Approach*
   6th edition, Jim Kurose, Keith Ross, Pearson, 2017

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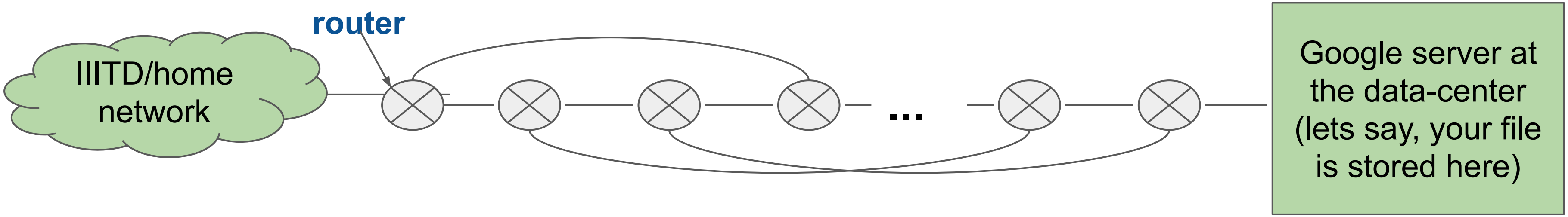
Brainstorming task (Take 2 minutes)



You want to upload a 10GB file to your google drive. Routers direct your data towards the destination server.

Requirements & Assumptions:

- The file is broken down into smaller pieces (say, 1000B). Let's call each piece, a packet
- Routers decide the path (next router) based on policies
  - *Could you think of appropriate policies?*

Think of all possible challenges that you will have to address for successful file upload

Challenges in end-to-end communication

- Addressing
  - Identification of end-devices (source, destination) and routers
- Routing
  - Compute possible paths to reach destination, apply policies while assigning paths
- Managing errors
  - Detect errors
  - Correct errors or retransmit packets??
- Managing faults and failures
  - Routers may crash or malfunction
  - Employ self-healing mechanisms
- Managing congestion
  - long packet queues at routers due to
    - High demand
    - Fewer resources assigned
    - Bad resource management
- *Many more ...*

Protocols, Internet standards, Communication models

- Protocols
  - Defines
    - Format and order of messages exchanged between entities
    - Actions taken after message reception/transmission
  - Example: TCP, HTTP, IP, …
- Internet standards
  - RFC: Request For Comments
  - IETF: Internet Engineering Task Force
- Communication models OR The Network(ing) Stack
  - The TCP/IP model or the Internet model
  - The OSI reference model

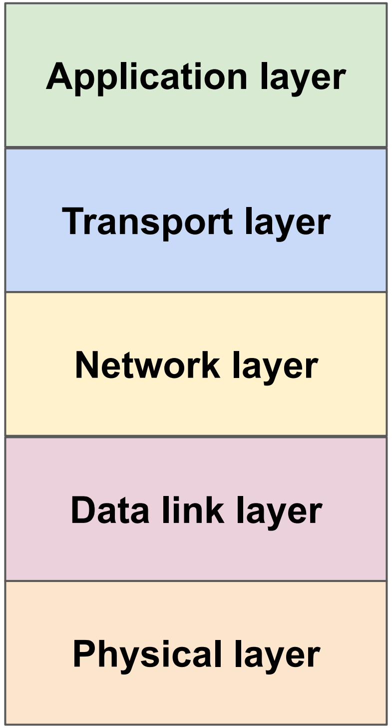
Terms you should know by now

- Broadcast, Multicast, Unicast, Anycast
- Connection-oriented vs. Connectionless
- Reliable vs. Unreliable message delivery
- Stateful vs. Stateless

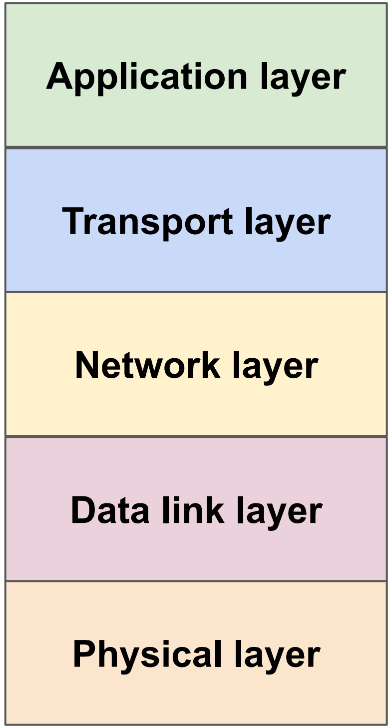
Layering concept

- Should we bundle all networking functions as one BIG software module?
  - The communication model comprises of multiple logical layers
  - Each layer handled several networking stack functions
- A layer at one host talks to a peer at the other host using a set of protocols followed by both the hosts
- A layer provides “services” and “abstractions” that the higher layer uses
  - In our earlier example, the HTTP protocol uses the services of TCP to transport the data to the web server
- Layering, why?
  - changes in one module will not affect the others
  - Easier to manage the software that implements a smaller set of coherent functions

The TCP/IP model (a.k.a., The Internet model)

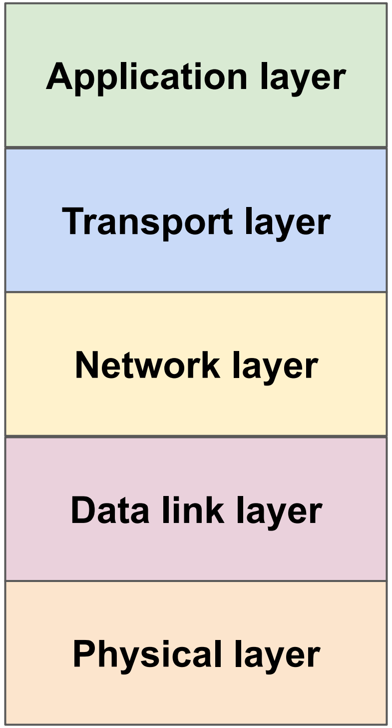


TCP/IP model: Functions of each layer



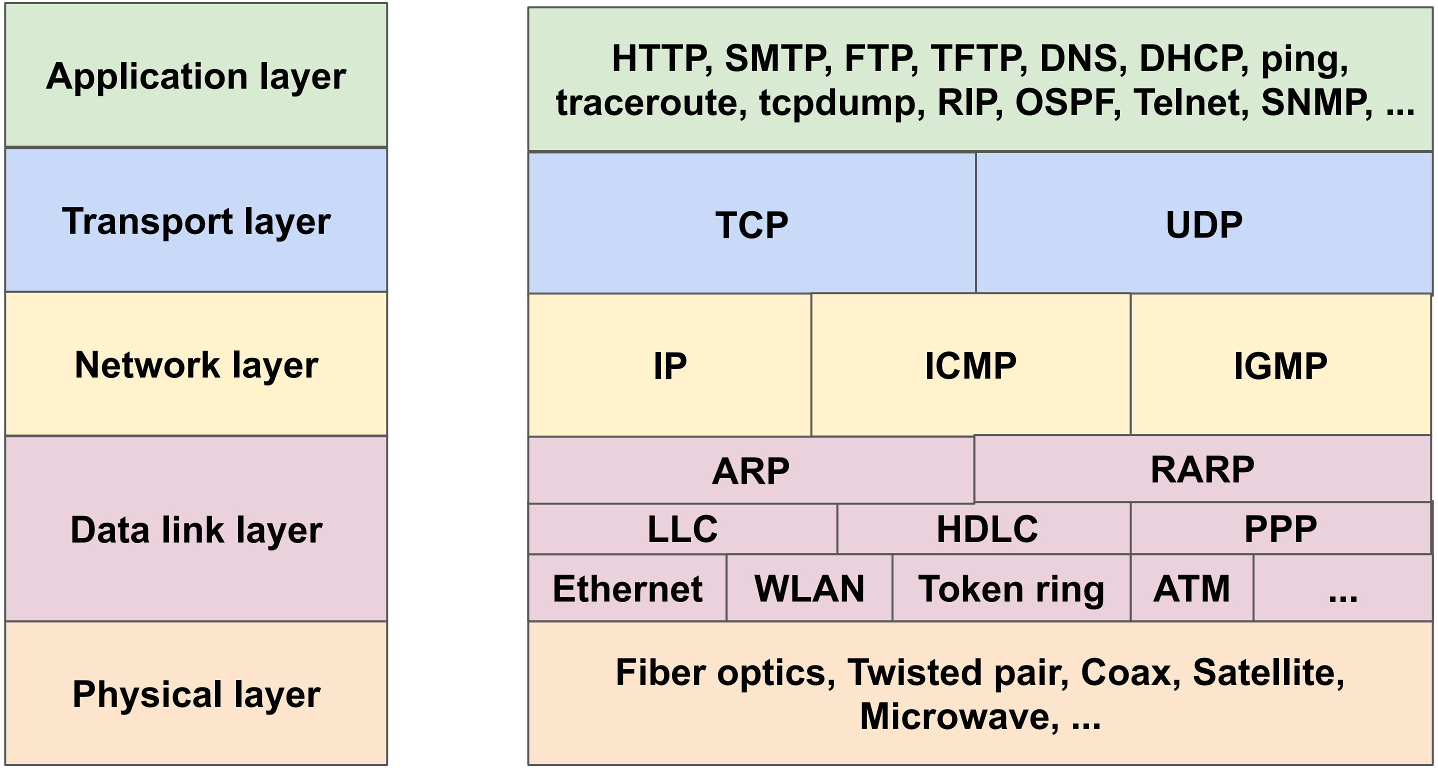
- Application layer
  - Concerned with application-level semantics
  - For example, a web server holds all the web pages in the disk/memory or computes dynamically and serves them when an HTTP GET request arrives
- Transport layer
  - Transports application layer “messages” from the source process to the destination process (using sockets)
  - Divides application layer “messages” into “segments”
  - TCP, connection-oriented & reliable
    - Flow control
    - Error control
    - Congestion control
  - UDP, connectionless & unreliable
  - RTP, used for real time services like VOIP

TCP/IP model: Functions of each layer (contd.)



- Network layer
  - Packs “segments” into network layer “packets”
  - Delivers packets from source to destination (using IP address)
  - Routing
  - Fragmentation & reassembly
- Data link layer
  - Packs “packets” into data link layer “frames”
  - Delivers frames from point to point within the same network, i.e., same broadcast domain (using MAC address)
  - Access control
  - Flow control
  - Error control
- Physical layer
  - Transports “frames” as physical layer “bits” over the physical links

The TCP/IP protocol stack



TCP/IP model: Encapsulation & decapsulation

Open Systems Interconnection (OSI) model

Application layer
Presentation layer
Session layer
Transport layer
Network layer
Data link layer
Physical layer

- Presentation layer
  - Allows applications to interpret data
  - E.g., encryption, compression, …
- Session layer
  - Synchronization
  - Checkpointing
  - Recovery of data exchanged
- Missing in the Internet stack??
  - Services needed?
    - Should be implemented by application

OSI model is not implemented

Example to understand layering concept

What happens when you open a web page from a computer by typing in the URL?

- URL gets resolved into IP address of the corresponding web server (DNS)
- Browser opens connection (TCP) with the web server
- Browser sends web page request (GET request of HTTP)
    - Browser may send multiple GET requests for different web page objects, ***may open multiple TCP connections***
- The data from the browser must be ***routed via multiple routers*** until it reaches the server’s IP address.
- Server sends some HTML content, browser displays